Part 03:
AI Model Building

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AI Model Building

Topics:
- AI Model building and performance metric
- Responsible AI practices in terms of auditability, reproducibility and documentation
- Human involvement in AI models

Learning Outcomes:
- Assess appropriate AI model building process and use appropriate AI model performance metrics
- Know how to achieve better AI project transparency in terms of auditability, reproducibility and documentation
- Know how to assess the required level of human intervention

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Responsible AI Model Building and Performance Metric Evaluation:

Ensuring AI Model Built is Robust, Reliable and Performing the way it should be.

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Considerations During AI Model Building

- Data splits
- Which model or algorithms to use based on problem statement (e.g., classification, regression, optimisation, etc)
- Architecture of the model
- Appropriate performance evaluation metrics
- Choosing between the different models based on output results

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Data Preparation for Performance Evaluation & Reliability

A data set can be divided into the following components:

- Training Dataset

  Used for development of the model during the training phase

- Testing Dataset (hold-out data set)

  Used to evaluate how well a model performs on unseen data

- Validation Dataset (May not always be necessary)
  - Not used for training the model
  - Trained/fitted model is used to predict the values in the validation dataset.
  - Used to provide an estimate of model readiness and for tuning of parameters, e.g., as a stopping criterion.

Sometimes, the term test dataset and validation dataset are used interchangeably.

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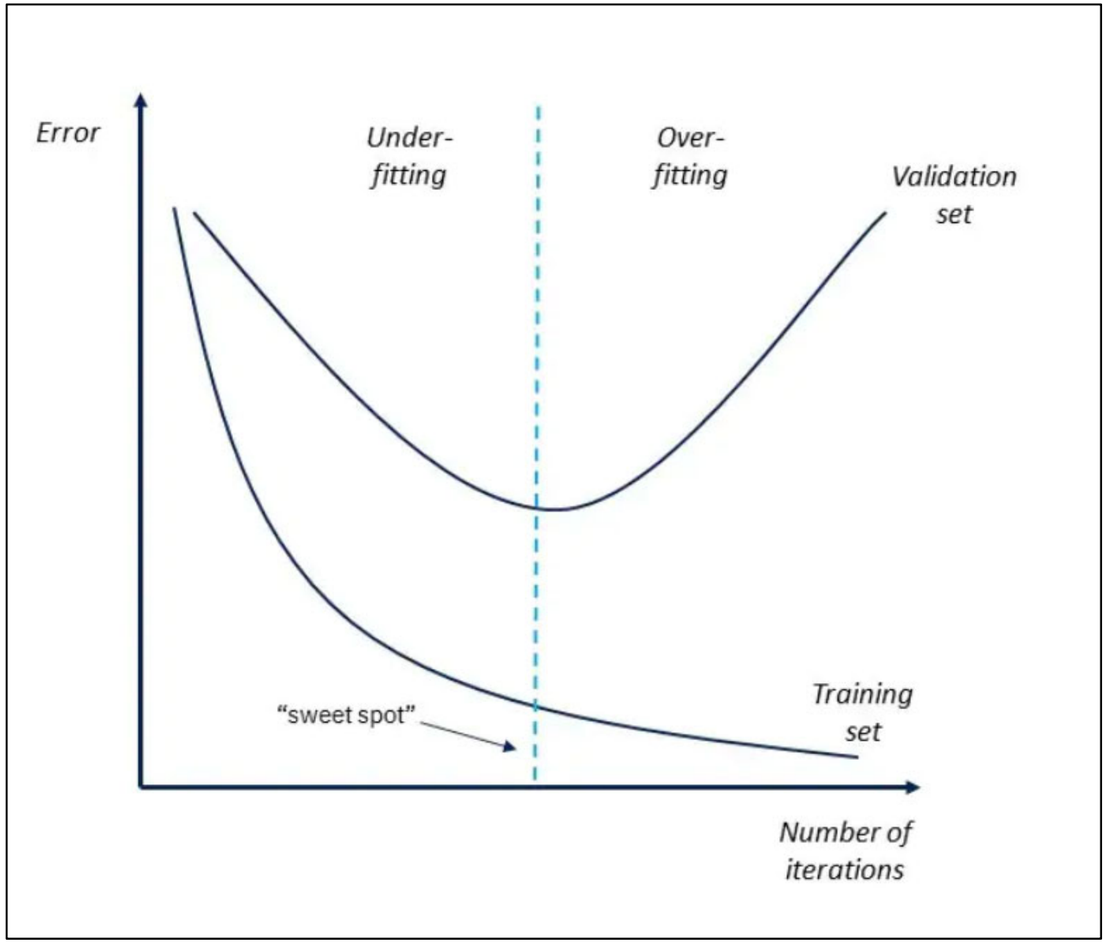
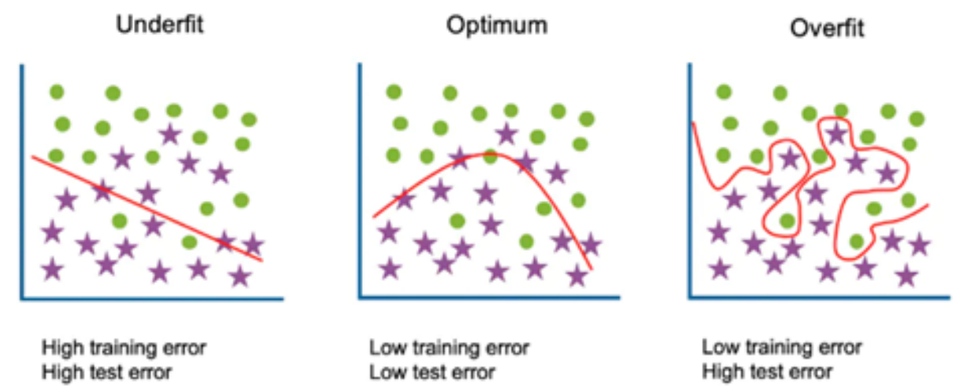
Data Preparation for Performance Evaluation & Reliability



Use of Validation Dataset



https://www.ibm.com/cloud/learn/overfitting

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Data Preparation for Performance Evaluation & Reliability

When the AI model is verified on the Train/Test split data, a single point of evaluation is obtained from the Train/Test split data.

- Is this enough to conclude that the model is reliable and robust enough?
- Will it perform similarly on the different segments of data?
- How to maximise the usage of the limited number of data that is available?

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Data Preparation for Performance Evaluation & Reliability

For Model Reliability & Robustness Evaluation

- *K*-fold cross validation may be used to ensure model performs well across different segments of the data.

- Random selection of training & testing datasets for *K* times.

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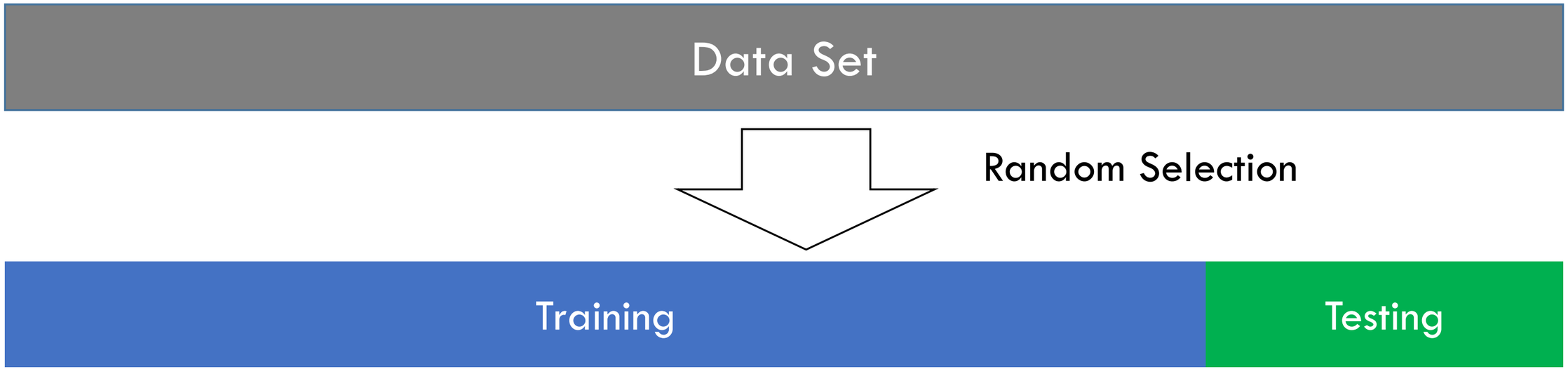
Data Preparation for Performance Evaluation & Reliability



Repeat this *K* times

Take the average of the *K* runs. Present the average results with the standard deviation.

Accuracy	AI Model
Training	80.5 (± 0.3)
Testing	77.8 (± 0.25)

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Hyperparameters Tuning

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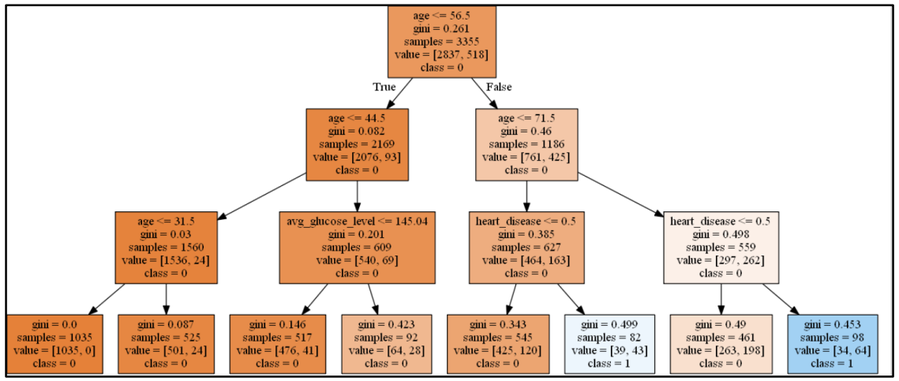
Hyperparameters Tuning

Generally, AI Models will require hyperparameters tuning.
Thus it is common to compare results across the same AI algorithm with different hyperparameters



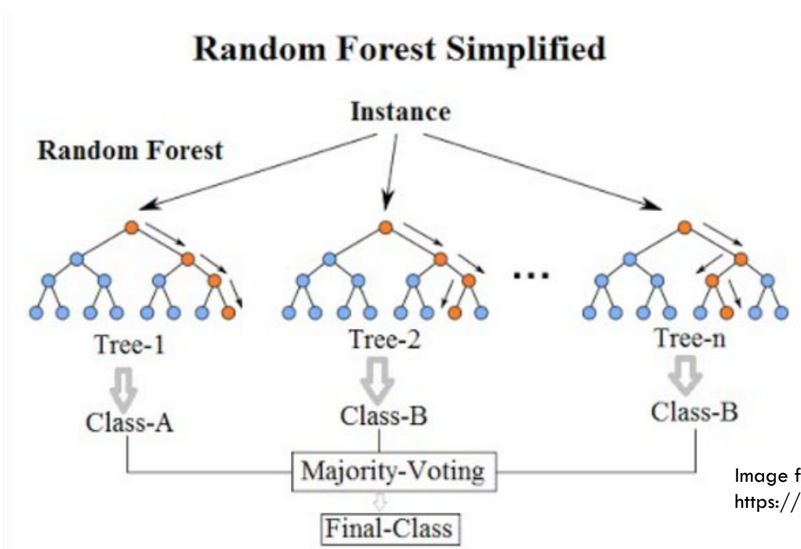
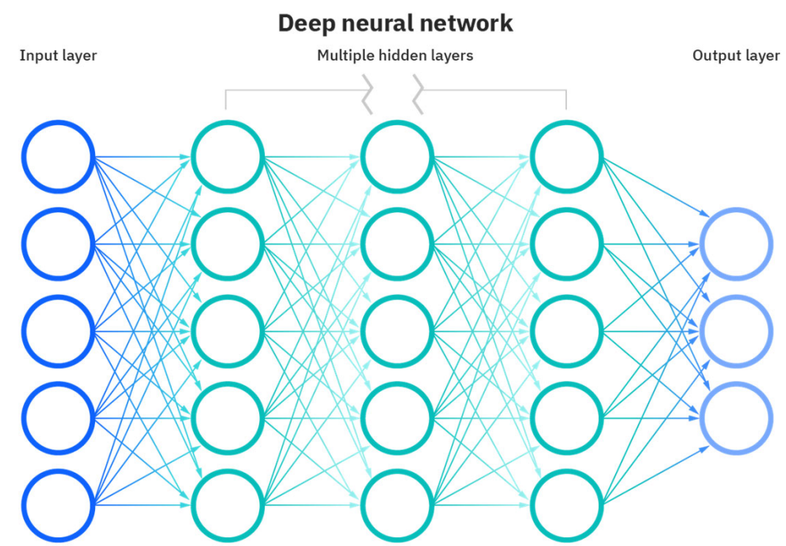


Image from:
https://en.wikipedia.org/wiki/Random_forest



https://www.ibm.com/cloud/blog/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks

If A_{11} and A_{12} and …. A_{1n} *then* C_1

else if A_{21} and A_{22} and …. A_{2n} *then* C_2

……

else $C_{general}$

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Performance Evaluation, Reproducibility & Reliability

- Take the average of the *K* runs. Present the average results with the standard deviation.
- Compare across the various models

Accuracy	Model 1a	Model 1b	Model2a	Model2b	Model3
Training	80.5 (± 0.3)	83.5 (± 2.3)	82.5 (± 0.35)	81.5 (± 0.3)	83.4 (± 1.3)
Testing	77.8 (± 0.25)	78.5 (± 3.8)	79.8 (± 0.22)	75.8 (± 0.15)	80.7 (± 2.8)

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Evaluation Metrics – Example Classification Accuracy

- Accuracy $= \frac{c}{n} \times 100\%$

  Where
  - c is the total number of correctly classified samples
  - n is the total number of samples

- Is accuracy the only evaluation metric?

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Confusion Matrix

We can further analyze the model performance by breaking down the results.

Consider a Binary Classification Problem

		Predicted Values	
		Negative	Positive
Actual Values	Negative	True Negative (TN)	False Positive (FP)
	Positive	False Negative (FN)	True Positive (TP)

Correct classification is achieved when:

- True Negative: When the actual value is negative(e.g., 0) and the predicted value is also negative (0)
- True Positive: When the actual value is positive(e.g., 1) and the predicted value is also positive (1)

Error will occur in 2 scenarios:

- False Positive: When the actual value is negative(0) and the predicted value is positive (1) (type I)
- False Negative: When the actual value is positive(1) and the predicted value is negative (0) (type II)

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Confusion Matrix

		Predicted Values	
		Negative	Positive
Actual Values	Negative	True Negative (TN)	False Positive (FP)
	Positive	False Negative (FN)	True Positive (TP)

- Accuracy = TP+TN/(TP+TN+FP+FN) %

- Precision/Hit Rate (of Event) = TP/(TP+FP)
Example: How many customers predicted as defaulter really default on the payment

- True Negative Rate/**Specificity** = TN/(TN+FP)
Example: **Percentage of non–defaulter that are correctly predicted** as not defaulting on credit card payment

- Recall/True Positive Rate/**Sensitivity** = TP/(TP+FN)
Example: **Percentage of credit card defaulters that are correctly predicted** as defaulters

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Try it Out

Discuss

Which model will you choose?

- **Medical Diagnosis of Cancer**
- **Prediction of Credit Defaulters**

		Predicted Values	
		Negative	Positive
Actual Values	Negative	765	55
	Positive	154	26

Accuracy = (765+26)/1000 = 79.1%

Specificity = 765/(765+55) = 0.933

Recall / sensitivity= 26/(26+154) = 0.14

		Predicted Values	
		Negative	Positive
Actual Values	Negative	605	215
	Positive	42	138

Accuracy = 605+138 /1000 = 74.3%

Specificity = 605/(605+215) = 0.738

Recall/sensitivity = 138/(42+138) = 0.767

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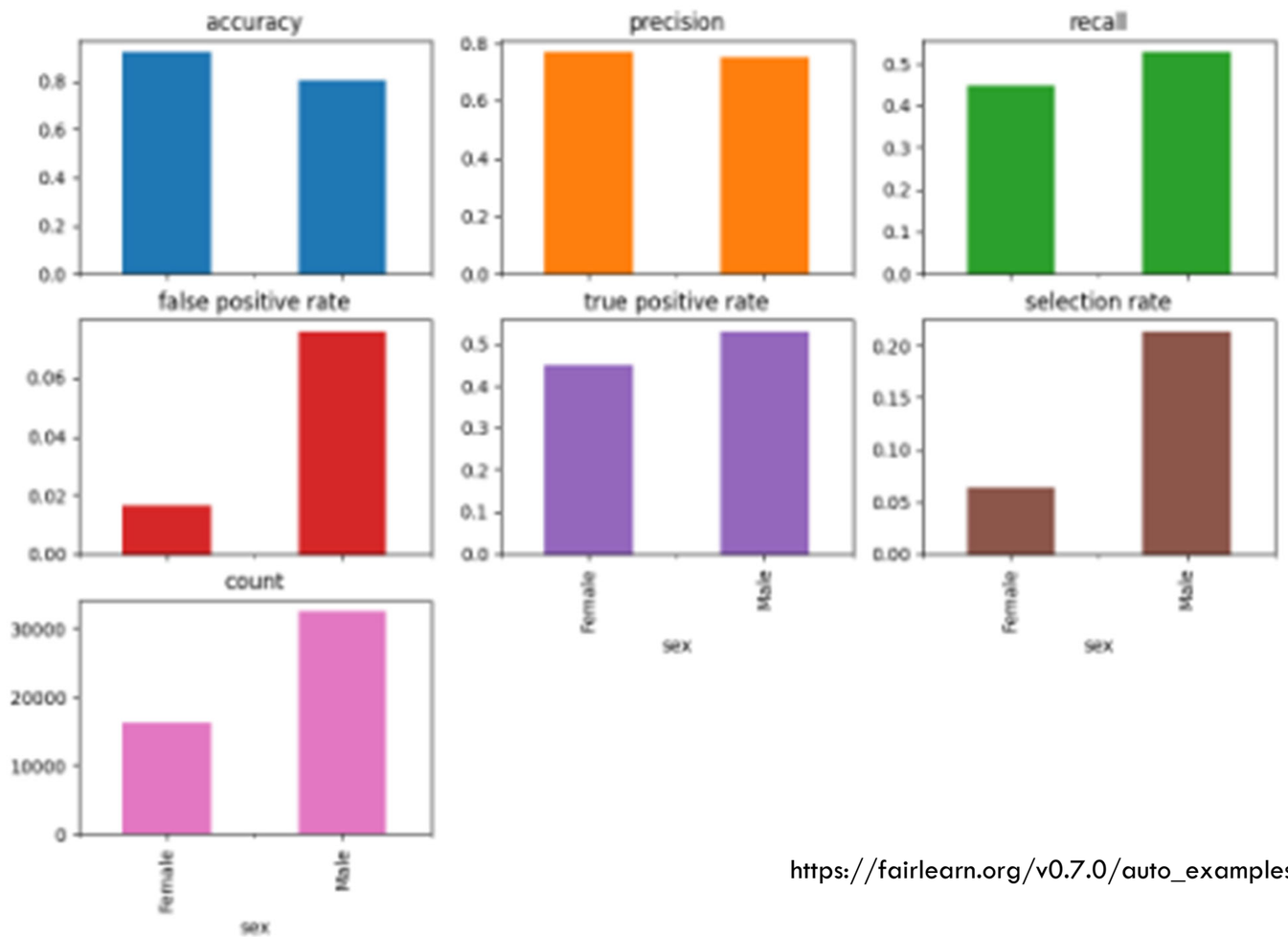
To Ensure Fairness across Subgroups, e.g., Gender, Ethnic, Age….
Looking into the Subgroups



https://fairlearn.org/v0.7.0/auto_examples/plot_quickstart.html

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Other metrics

1. Statistical Parity/Demographic Parity: Ensures that individuals from **different groups** have the same probability of receiving a **positive outcome.**

2. Equal Opportunity: Ensures that **qualified** individuals from **different groups** have the same probability of receiving a **positive outcome.**

3. Equalized Odds: Require that the **true positive rate** and **false positive rate** are the same for **different groups.**

4. Predictive Parity: Ensures that the predicted positive outcome has the **same precision across different groups.**

5. Treatment Equality: Focuses on balancing the ratio of **false positives** to **false negatives across different groups.**

Extracted from https://shelf.io/blog/fairness-metrics-in-ai/

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Responsible AI Practices in Terms of Auditability, Reproducibility and Documentation

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Audit to Built Trust

Value of Audit

Building confidence and trust

Audit is able to help organisation built trust and confidence in the organisation.

Building confidence and trust in capital markets

Audit is vital to trust in financial information, and it's this trust that turns the wheels of the capital markets system. It is the independence that auditors bring, the objective scrutiny, which gives the business community confidence in the numbers. As a profession, we need the respect and cooperation of management, but we must also respectfully challenge what we are given and make our own independent assessments. In discussions with our stakeholders and investors, there was also a recurring recognition that the audit needs to evolve with the changing times if it is to meet investor needs. A number of participants in the discussions raised the concern that the audit's focus is on historical information – a largely backward-looking view. But what investors are increasingly looking for is assurance over prospective information. Providing broader assurance, including over more forward-looking information, is one of the key challenges in many participants' view facing the audit.

https://home.kpmg/xx/en/home/insights/2018/04/building-confidence-and-trust-in-capital-markets.html

Likewise, Audit for AI can help to build trust and confidence in the AI

KPMG

Why AI must be included in audits

By Paul Holland, Shamus Rae and Paul Taylor

https://assets.kpmg/content/dam/kpmg/uk/pdf/2018/06/why-ai-must-be-included-in-audits.PDF

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AI Audit

- AI Audit strengthens Trust, Confidence, Transparency, Accountability and Responsibility
- Audit reports serves as an independent and un-biased view on the AI model developed and provides certain level of assurance to all stakeholders.
- Verify whether the AI model is performing as expected and that the output results is within the expected range, given the assumptions and constraints provided. This is important as many AI systems function as black box.
- Audit helps to determine whether the project team is providing a true and fair representation of what they have presented and stated as doing.
- It also determine whether the AI project team has followed and implemented the responsible AI practices and principles, or has complied to the required organisation governance or legal regulations.

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AI Audit

- Reports may detect & reveal weakness in AI model development process and provide valuable feedback to management and project team.

- AI Model audit could be done at periodic basis by internal or external auditors. This will also depends on whether the organisation has the expertise. If not may consider training the auditors with the relevant expertise. Pre-deployment audits are also recommended.

- Audit should be performed on the various steps on AI model building process.

- The audit report reflects the overall position of where the organisation stands in responsible AI practices and principles

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AI Model Documentation

Documentation serves multiple purposes and one key factor is to allow audit process to be performed smoothly.

In terms of Responsible AI aspects, documentation is important and can serve to address the following questions if being raised:

- Is there any bias in data selection for training?
- Are there any ethical issues in the modelling(e.g., one that causes harm)?
- Is there a structure in place for data access, who can access and modify the data and model?
- Does the model always perform within a certain level of confidence and similar way?
- What is the robustness and accuracy of algorithms?

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AI Model Documentation

- Other key reasons for documentation would be to facilitate knowledge transfer, easy communication and collaboration among different teams, serves as a trail of the events, etc.

- Documentation also allows reproducibility of AI models and the processed data. E.g., code comments, what transformation were being performed on the data and the reason, etc.

- Act as safeguards and provide evidence in the event of disputes.

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Data Documentation & Data Lineage

Data lineage documentation is important as it provides the ability to trace back to the root cause in the event of error, enable reproducibility, sets in place accountability, etc.

- Context of data
- Source & origin of data
- Date data is collected/retrieved
- Period of data
- Data dictionary (feature name, description, etc)
- Transformation to the data
- Log of data movement

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Reproducibility of AI Model

- Previously discussed reproducibility in terms of AI Models **results**, i.e., model robustness.

- How about the re-production of AI **Models**?

- Can be done only if you know the working mechanism and have the source code with proper documentation.

- Implement **good coding standards & practices** for both **reproducibility of AI model as well as model results**

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Types of Documentation
Responsible AI Model Building – For Audit, Compliance, Reproducibility, Accountability

Discuss

What are the types of documentation that you practice in your organisation?

What are those documentation relevant to AI model building?

Any other documentation that is important to AI model building process?

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AI Model Building Confidentiality

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AI Confidentiality

From AI User Perspective

- Data collection to have consent of user and user is aware of it
- Ability to opt in or opt out
- Trust that data will not be stored (or is to be deleted) after some time
- Usage of data is for specific purpose and will benefit the user
- Expect process to be well explained when required, how the data is being used and who has authorized access, etc.

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AI Confidentiality

From AI Owner Perspective

- Gain trust from the users of AI or data subjects that their confidentiality is protected
- AI model is a valuable asset and needs to be protected
- Performs AI Audit to the extent that no confidential info is being revealed
- Protect AI algorithms and details, to prevent leakage to market competitors and to prevent AI algorithms from being exploited and attacked.
- Reveal only at a need-to-know info basis to the respective parties

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Responsible AI:
How much Human Involvement in AI Models?

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Human Intervention

Can AI be totally autonomous, semi-autonomous or requires human intervention?

Human In The Loop

- Actively involved
- Makes the final decision
- Full control
- AI just as a complement

Human Over The Loop

- Overseas AI function
- Choose or overwrite when required

Human Out Of The Loop

- Not involved
- AI fronts the customer/user

IMDA: Model Artificial Intelligence Governance Framework Second Edition

Human Intervention

Human In The Loop

Human Over The Loop

Human Out Of The Loop

Consider the 3 buttons

Which button to press and in which situation?

What are the considerations to the extend of human intervention in AI?

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youtube.com/@nus-iss

Thank you

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